PRE-READ · SUBMARINE CABLES · CHINA A-SHARES

# Ningbo Orient

## Wires & Cables Co. Ltd. · 603606 CH

Everything you need to know — in plain English — *before* you open the dashboard. What submarine cables actually are. Why this is one of the most defensible industrial businesses in the world. And why China's offshore-wind buildout has put Ningbo Orient at the centre of a multi-year boom.

| RMB 42 bn | 33% | RMB 18.4 bn | +26% | RMB 78 |
|---|---|---|---|---|
| Market cap | China submarine cable share | Order backlog (Apr 26) | FY25 net profit growth | UBS price target |

Reading time: ~12 minutes

Companion to the interactive dashboard at

**juankhye.github.io/ningbo-orient-603606-dashboard**

SECTION 00

# What's in this document

SECTION 01

# The one-paragraph pitch

Ningbo Orient is a Chinese industrial company that makes the giant electrical cables lying on the seafloor — the cables that carry electricity from offshore wind farms back to land. It is one of just three companies that dominate this business in China, and one of fewer than ten companies in the entire world that can manufacture the highest-voltage version of these cables. The world is now building offshore wind farms at the fastest rate in history: China alone plans to install 100 gigawatts of offshore wind by 2030 — roughly five times today's installed base. Every one of those wind farms needs Orient's cables to connect to the grid. The order book is full through 2027, the balance sheet is debt-free, and the stock trades at a discount to both its own history and its peers.

**WHY THIS MATTERS**

If you understand nothing else: this is an industrial business that economically behaves more like defence contracting than like a normal manufacturer. The customer cares about reliability above price. New competitors take a decade to break in. Existing players defend their share fiercely. That is what makes Orient interesting — not the topline growth alone, but the structural reasons that growth flows through to durable profits.

## In five sentences

**1. Structural duopoly with a moat dug by physics.** Cables fail catastrophically when wrong, so developers stick to a handful of qualified suppliers. China's top-3 control 87% of the market.

**2. Order book locks in revenue through 2027.** RMB 18.4 billion of signed contracts — roughly 1.7 years of trailing revenue, weighted toward high-margin 500kV product.

**3. China policy mandates the demand.** 15th Five-Year Plan targets 100GW of installed offshore wind by 2030, implying ~10GW per year vs 5-7GW in the prior plan period.

**4. Europe is the next leg, validated by delivery.** Orient delivered the UK's Inch Cape project in 2025 — proving Chinese cable meets European specs at 10% lower price.

**5. Cheap on its own growth and vs. peers.** Trades at 18x 2027 earnings on 25% earnings growth, below the peer average and below its own 27x historical multiple.

SECTION 02

# What is a submarine cable, really?

Picture a thick steel-wrapped rope, about as wide as your forearm, weighing 100 tonnes per kilometre, lying on the seabed and stretching as long as 200 kilometres. Inside that rope: three copper conductors carrying enough electricity to power a million homes, wrapped in plastic insulation, wrapped in lead to keep seawater out, wrapped in steel wires to survive being dragged by anchors and fishing trawlers.

That is a submarine cable.

When you build a wind farm in the ocean, the cable is what carries the electricity those turbines generate back to a substation on shore. Without the cable, the entire wind farm is just an expensive ornament spinning uselessly in the water.

## Why this matters

Submarine cables look boring. They are not. A single 500-kilovolt HVDC submarine cable can cost €5–10 million *per kilometre*. A typical offshore wind farm needs 50–200 kilometres of them. That puts cable spending into the hundreds of millions of dollars per project — and submarine cables represent 8–13% of total offshore wind project cost, comparable to the cost of the turbines themselves.

## Now the kicker

Cables fail. Cables fail *catastrophically*. When a cable fails, the entire wind farm — which might have cost a billion dollars to build — goes offline. Repairs require specialised ships that are in short supply globally. A single cable failure can shut down a wind farm for six months.

**THE SINGLE MOST IMPORTANT NUMBER IN THIS INDUSTRY**

83% of all insurance claims in offshore wind trace back to cable failures. That one statistic explains everything that follows: why customers do not shop on price, why suppliers need track records before they get qualified, why banks demand pre-approved vendors, and why the global market is locked up by fewer than ten companies.

SECTION 03

# Why this is a weirdly defensible business

If 83% of your project's insurance risk lives in a single component, you do not choose that component based on price. You choose it based on track record. You choose suppliers your bank trusts. You choose suppliers your insurer trusts. You choose suppliers your grid regulator has pre-approved.

This is why submarine cables are nothing like a normal industrial product. There are five reasons new competitors cannot break in:

1. **Manufacturing is genuinely hard.** Making a 500-kilovolt cable that survives 25 years underwater takes specialised chemical engineering (cross-linked polyethylene insulation), proprietary high-voltage testing equipment, and a manufacturing process that costs hundreds of millions of dollars to set up. Fewer than 10 companies on Earth can do it.

2. **You need a deep-water port.** Submarine cables are spooled onto cable-laying ships at the factory. Those ships are huge. They need deep-water access. There are only a handful of suitable ports in China, and getting permission to build a new one takes years — not months.

3. **You need your own ships.** Cable-laying vessels are scarce globally. If you don't own the ship, you can't deliver the project. Orient owns three of these ships, including an 18,000-tonne transport platform that ships completed cables straight to the UK.

4. **You need to be on the "approved supplier" list.** China's National Grid maintains a closed roster of just six companies that are approved to bid on high-voltage interconnection projects. Orient is one of those six. Banks won't finance a project that uses an unlisted supplier.

5. **You need a track record.** Even if you somehow built the manufacturing, the port, and the ships — wind farm developers won't risk their billion-dollar project on a supplier that's never delivered one before. New entrants need a customer willing to take a leap of faith on day one. That customer doesn't exist.

**The result:** China's domestic submarine cable market is shared by three companies (Zhongtian 37%, Ningbo Orient 33%, Hengtong 17%) that together control 87% of the market. Globally the picture is similar — a tiny club of qualified suppliers.

**AN ANALOGY**

This business is closer to defence contracting or commercial aviation (Boeing-Airbus) than to traditional industrial manufacturing. The customer cares about reliability above price. The barriers to entry are technological and regulatory rather than capital-only. Existing players defend their share fiercely.

SECTION 04

# What Ningbo Orient actually does

The company sells three things:

## 1. Submarine and high-voltage cables — the crown jewel

These are the multi-million-dollar cables described in Section 2. Voltage classes from 35kV array cables (which connect turbines within a wind farm) through 220kV and 500kV export cables (which carry power back to shore) to ±525kV HVDC cables (for the longest distances). This segment is 50% of revenue and generates a 33% gross margin. It is where the moat lives, and where every story about this company has to start.

## 2. Land cables — the workhorse

Medium- and low-voltage power and data cables used by utilities, telecoms, real estate, rail networks. This is a commodity business — many competitors, ~9% gross margin. Orient does it because it utilises manufacturing capacity and serves big state-owned customers, but nobody owns this stock for the land-cable business.

## 3. Marine engineering services — the differentiator

This is the team that designs the cable route, lays the cable at sea, joints the cables together, tests them, and provides ongoing maintenance. Only ~7% of revenue, but it is the reason customers pick Orient over a pure cable maker. Increasingly, customers buy a "turnkey" package — cable plus installation plus maintenance — rather than just the cable. Orient is one of the few companies that can offer the whole package.

**MENTAL MODEL**

Orient is not a cable manufacturer competing on price. It is an infrastructure provider competing on reliability. It just happens that the infrastructure is shaped like a cable.

SECTION 05

# Why now? The two big tailwinds

Two megatrends are pushing demand for submarine cables harder than at any moment in history. Both run for at least five years.

## Tailwind 1 — China's 15th Five-Year Plan

China's central government has just published its 2026–2030 plan, with a target of 100 gigawatts of cumulative offshore wind capacity by 2030. Roughly 49 GW is installed today. To hit the target, China needs to install around 10 GW per year for five years. That's a 39% step-up from the prior plan period. The pipeline already supports it — over 50 GW of projects have received initial approval, and 8.4 GW received final approval in 2025 alone.

Translation: a structural, government-backed demand tailwind running through 2030. Submarine cable suppliers can plan production years in advance because the demand is locked in by policy, not market sentiment.

## Tailwind 2 — Europe's capacity crunch

Here it gets interesting. Europe is also accelerating offshore wind aggressively — the Hamburg Declaration targets 100 GW in the North Sea by 2050; the UK's AR7 auction allocated 8.4 GW; France ran a record 10 GW tender. But European cable manufacturers (Prysmian, Nexans, NKT) are sold out through 2027–2028. They cannot add capacity quickly enough.

Meanwhile, Chinese suppliers including Orient have spare capacity, sell at ~10% below European prices (even after shipping costs), and now have a track record: Orient delivered cable for the UK's Inch Cape project in 2025, proving it can meet European technical standards.

**TRANSLATION**

Europe has to buy Chinese cable for the next few years whether it wants to or not. The only question is how much before geopolitical concerns (the UK recently blocked another Chinese wind company on national-security grounds) start to kick in. For Orient, this means overseas revenue grows from 11.6% of total in 2025 to 15–20% by 2030 — and at higher margins than domestic.

SECTION 06

# The numbers in plain English

**Revenue (the top line).** RMB 10.8 billion in 2025. UBS forecasts this grows to RMB 28 billion by 2030 — a 21% annual growth rate sustained over five years. That kind of growth is rare for an industrial company.

**Gross margin (how much they keep per dollar of revenue).** 22% in 2025, projected to expand to 26% by 2030. The expansion comes from mix shift: more revenue from high-margin 500kV submarine cable, less from low-margin land cable.

**Net profit (the bottom line).** RMB 1.27 billion in 2025, up 26% year-on-year. Q1 2026 was even better — up 32% year-on-year. The trend is real.

**Order book (the visibility).** RMB 18.4 billion as of April 2026 — roughly 1.7 years of revenue already committed. Heavily weighted toward high-margin submarine and HV cable. This is not a "hope it sells" story; customers have already paid deposits.

**Balance sheet (the resilience).** Net cash position of RMB 2.2 billion. Debt-to-equity just 9.8%. This company can fund its expansion plans entirely from internal cash flow — no need to dilute shareholders or take on risky debt.

**Valuation (what it costs to own).** Trades at roughly 22.6× next year's earnings and 18× the year after's. For a company growing earnings ~25% per year, this is cheap. Peer companies trade at 23–27×. The stock's own historical average is 27×. So you're paying below average for above-average growth.

## Analyst target prices

| Broker | Rating | Price target (RMB) | Upside |
|---|---|---|---|
| UBS | Buy | 78.00 | **+28%** |
| Morgan Stanley | Overweight | 69.63 | **+14%** |
| Huatai | Overweight | 68.38 | **+12%** |
| Credit Suisse | Outperform | 69.50 | **+14%** |

Consensus is bullish, with target prices about 14–28% above where the stock trades today.

SECTION 07

# What could break the story

Every investment thesis has things that could kill it. Three risks are worth watching closely. Several smaller risks are real but secondary.

## Risk 1 — Copper

Submarine cables are basically copper conductors with insulation around them. Copper is 40–50% of the raw-material cost. If copper prices spike 30% and Orient can't pass that through to customers, margins compress hard. The company hedges 100% of its submarine cable orders, which helps. But this is the single biggest input-cost risk.

## Risk 2 — Project delays

Offshore wind installation depends on weather, vessel availability, sea-use permits, military airspace clearance, and port logistics. Any of these can delay deliveries. When they do, Orient's revenue gets lumpy: Q2 2025 saw a 50% profit drop because Q1 deliveries pushed into Q3. This is the kind of quarterly volatility that scares short-term investors but doesn't change the long-term thesis.

## Risk 3 — A European ban

The UK government recently blocked a Chinese wind turbine maker (Mingyang) from building a factory in Scotland, citing national security. Submarine cables are arguably more sensitive infrastructure than turbines. If the UK or EU were to impose a similar ban on Chinese subsea cable imports, the 15–20% overseas revenue target would compress. The base case treats this as low-probability, but it is a real tail risk.

## Lesser risks worth knowing about

- Capacity expansion by competitors Zhongtian and Hengtong could compress industry prices in a post-subsidy era.
- "Sea-use" bottlenecks (military airspace, port logistics, vessel availability) could cap installations below plan targets.
- Floating offshore wind technology could disappoint on timing, deferring the dynamic-cable opportunity.
- EU local-content rules could erode the 10% Chinese price advantage even without an outright ban.

SECTION 08

# The operational proof

Everything up to here is the story — the structural reasons this looks like a good business. Everything in this section is the contract-level evidence that confirms the story is real. Balance-sheet items. Customer-by-customer revenue. Per-kilometre unit economics. These are the numbers you can put on a memo without using the word "should."

## Take-or-pay is structurally binding

The single most important piece of evidence in this entire pitch lives in one line of Orient's 2025 annual report. Under the breakdown of performance obligations, the company explicitly reports the "expected amount to be refunded to customers" for both domestic and overseas submarine cable products as **RMB 0.00**. The same zero appears in the 2024 annual report.

**WHY THIS MATTERS**

When a wind-farm developer signs a cable contract with Orient, they pay a cash deposit upfront to reserve manufacturing capacity. If they cancel, they forfeit the deposit. The annual report's zero-refund disclosure is empirical take-or-pay — the same kind of evidence pharma analysts look for in companies like Stevanato Group, where €132 million of customer prepayments validated the contractual structure.

How much money is sitting on the balance sheet under this regime? Contract liabilities surged from RMB 1.67 billion (Q2 2025) to RMB 2.37 billion (year-end 2025) to RMB 2.50 billion (Q1 2026) — a 50% increase across nine months. Performance bonds outstanding stand at CNY 415 million plus EUR 18 million — the operational scale of guarantees Orient has posted to customers.

**Backlog conversion math:** mid-2025 backlog of RMB 19.0 billion against full-year revenue guidance of ~RMB 10.0 billion implies roughly 52.6% conversion within twelve months. The other half flows into outer years.

## Customer concentration — decoded with honest uncertainty

Orient's annual report aggregates the top-5 customers as one anonymised line item — 59.6% of revenue collectively, 67.15% of accounts receivable + contract assets. Cross-referencing public tender data with developer 10-K disclosures decodes the top three:

| Rank | Customer | FY25 revenue | % of total | Driver |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 1 | China General Nuclear Power (CGN) | RMB 1.30 bn | 12.0% | Yangjiang Fanshi 2 & 3 |
| 2 | Inch Cape Offshore Ltd (UK · SSEN) | RMB 0.91 bn | 8.4% | UK 220kV export cable |
| 3 | China Energy Engineering (CEEC) | RMB 0.86 bn | 7.9% | Domestic EPC contracts |
| 4–5 | State Grid + Southern Power framework | ~RMB 3.4 bn | ~31.3% | Triangulated |

The gap between top-3 (28.3%) and disclosed top-5 (59.6%) implies the 4th and 5th positions are broad framework roll-ups — likely aggregated State Grid and Southern Power Grid provincial entities. Top-10 estimate: ~78% of revenue (75–85% band). Counterparty default risk is essentially zero because the anchors are state-owned, but execution concentration is high — a single Inch Cape slip moves 8.4% of revenue.

## Per-kilometre economics — the build-up

Submarine cable manufacturing is overwhelmingly a raw-material business. More than 90% of total manufacturing cost is direct raw materials. Per Credit Suisse's decomposition:

| Component | % of BOM | Note |
|---|---|---|
| Copper conductor | 37% | The single biggest input cost and the main margin risk |
| Other metals (aluminium, steel armour) | 28% | Standard procurement |
| Polymer (XLPE insulation) | 14% | High-voltage grades still imported from Europe |
| Plastics and other materials | 14% | |
| Manufacturing overhead | 5% | |
| Direct labour | 2% | |

Management has guided the ASP of a 500 kV submarine cable at approximately **RMB 14.0 million per kilometre**. A 220 kV cable carries an ASP roughly 30% lower (~RMB 10.8 M/km). A 66 kV inter-array cable runs ~RMB 1.5 M/km. ±525 kV HVDC commands the highest premium (~RMB 16 M/km, approximately 15% above 500 kV AC).

Margins peaked near **50%** in 2021 during the subsidised rush installation period. Management guides 30–40% normalized post-grid-parity, stabilising around 35% by 2025. Mid-2025 reported 25.0% on submarine cables — compressed during a delivery-delay quarter but expected to recover.

**THE DEFLATION RISK WORTH KNOWING ABOUT**

Bigger turbines reduce cable demand per GW. A 1 GW project with 6 MW turbines needs ~167 foundations and lots of inter-array cabling. The same 1 GW with 15 MW Vestas turbines (as used at Inch Cape) needs only 72 foundations and far less inter-array. Inch Cape consumed just ~154 km of export cable per GW — roughly half the 300 km/GW heuristic from France's AO10 tender. Even if installed GW meets target, kilometres-of-cable per GW may shrink. The bull case quietly assumes constant content-per-GW. It probably shouldn't.

SECTION 09

# Pipeline reality check

China's 100 GW target needs a haircut. Europe's supply deficit is real but narrower than the bull case implies. Here's the math.

## China — the 47% historical track-rate

During the 14th Five-Year Plan, China targeted 60 GW of cumulative offshore wind installations across 2021–2023. Only ~28 GW was actually installed during that window. That's a **47% historical track-rate** — implementation falling well short of plan. The reasons are operational, not policy: waterway coordination disputes, military airspace clearance, installation vessel shortages, port logistics, grid connection bottlenecks. A January 2025 Ministry of Natural Resources directive requiring new projects to be ≥30 km offshore or in waters >30 m deep further constrains the viable project zone.

Apply the same 47% track-rate to the 15FYP target of 100 GW cumulative by 2030: realistic installations come out at roughly **47 GW**, not 100. Industry channel research suggests a more bullish ~12 GW/year ceiling driven by capacity constraints — that lands at ~60 GW cumulative by 2030, still well below the headline target.

**Either way, the China demand story is real but smaller than the policy headline suggests.** Worth pricing that into the revenue model.

## Named pipeline projects driving 2026/2027 cable demand

| Project | Developer | Capacity / spec | Status | Expected COD |
|---|---|---|---|---|
| Yangjiang Qingzhou 5 & 7 | Three Gorges | ±500 kV | Under construction | 2026 |
| Yangjiang Fanshi 2 & 3 | CGN | 500 kV AC | Implementation | 2025/26 |
| Yangjiang Sanshandao 5 & 6 | CGN | 1.0 GW · 114.8km 500kV | Advancing | Q4 2026 |
| Shenneng Hainan CZ2 | Shenneng | 1.2 GW | Phase I connected Mar 25 | 2026/27 |
| Longyuan Hainan CZ8 | Longyuan | 500 MW | First monopile Oct 25 | 2027 |
| Datang Danzhou Phase II | Datang | 600 MW | Site work Dec 25 | 2027 |

There's a real pipeline. But not enough of it is fully funded, permitted, and vessel-slotted to support the 10 GW/year mandate. The realistic 2026–2030 install range is 30–60 GW cumulative.

## Europe — supply chain still tight; two flags to know about

European cable producers are sold out 4–6 years forward (Prysmian 5.2 years, NKT 5.2 years, Nexans 4.6 years). The structural undersupply of HVDC submarine cable is forecast at 1,500 km/year currently, widening to 2,500 km/year by 2030 even after the announced capacity builds. That's the window for Chinese overflow.

Capacity coming online: Prysmian doubles Pikkala (Finland) for 525 kV by 2026; Hellenic Cables added 50,000 tonnes/year at Corinth (Q1 2025); Sumitomo £350M UK plant mid-2026; LS Cable £1bn UK Tyne facility slipped from 2027 to >2028. The gap narrows but doesn't close before 2030.

## Flag 1 — Chinese suppliers' historical absence from European HVDC

Kepler Cheuvreux (June 2025) is direct on this: *"China's Hengtong and Orient Cable, alongside South Korean peers, remain largely absent from the European high-voltage and HVDC submarine cable markets."* Reasons cited: *"regulatory hurdles, technical standards, and deeply ingrained strategic and trust-related barriers among European transmission system operators."*

Translation: Inch Cape may be the exception, not the new rule. The 15–20% overseas revenue target leans heavily on Inch Cape becoming a reference for follow-on European wins, not on European TSOs broadly opening up. The conservative path is overseas revenue settling at 12–15% rather than 20%.

## Flag 2 — France AO10's 60% local-content wall

The French 10 GW tender — the largest single offshore wind procurement in European history, expected to award fall 2026 — incorporates Net-Zero Industry Act standards requiring ≥**60% European local supply chain** for submarine cables. That's not a tail risk; it's a hard exclusion on roughly 3,000 km of cable demand. Combined with EU FDI review thresholds (triggers above €100 million, forces JV with EU partner + IP sharing + 60–105 day review), Orient cannot economically build greenfield European capacity to bypass the wall.

**WHAT THIS MEANS FOR THE MODEL**

The bull case dashboard shows overseas revenue growing 11.6% → 15–20% by 2030. The Kepler "largely absent" quote and the France 60% wall together suggest the realistic landing is 12–15%, not 20%. Worth haircutting that line item in the revenue build.

SECTION 10

# Tripwires and disclosure gaps

Every thesis has things that, if observed, would prove it wrong. Below are the six specific monitorable thresholds that would flip this name short — each one observable in quarterly disclosure or public tender data. Below those: the metrics Orient doesn't disclose, plus the triangulation method for each.

## Six quantified tripwires

**1. Demand — Q1 or Q2 2026 China offshore wind bidding falls below 2 GW**

The 10 GW/year installation thesis requires healthy bidding flow. A sub-2 GW quarter implies installation pace closer to 14FYP (5–7 GW/year). Public bidding data; quarterly check.

**2. Margin — A 500 kV submarine cable tender awards at below RMB 12 million per km**

Management guides 14 M/km. A 14% ASP haircut at the highest-margin product would replay the 2021 turbine price collapse pattern and compress submarine gross margin from 35% to ~25%.

**3. Inventory — Submarine cable inventory grows more than 25% QoQ without matching revenue ramp**

Current 819 km finished-goods inventory at end-2025 (+147.6% YoY). Should convert to revenue through 2026. If it keeps building while revenue stays flat, downstream installation is stalling — replays the Q2 2025 pattern when net profit dropped 49.6% YoY.

**4. Competition — China top-3 oligopoly share falls below 80% on a trailing-four-quarter basis**

Currently 87% (ZTT 37%, Orient 33%, Hengtong 17%). A successful new entrant qualifying for State Grid framework on 500 kV+ products and winning provincial tenders would be the destabilising signal.

**5. Geopolitics — A named UK / EU national-security review action specifically targets Chinese submarine cable**

The Mingyang precedent (UK blocked Chinese turbine factory) and April 2024 EU anti-subsidy probe on Chinese turbines are pointing this way. A binding restriction naming subsea cable would haircut overseas revenue from the 15–20% target to single digits.

**6. Copper — Submarine gross margin falls below 28% for two consecutive quarters**

Copper is 37% of cable BOM. Orient hedges 100% of subsea orders via futures, but new orders need ASP renegotiation. A sustained copper spike unmatched by pass-through, plus competitive ASP pressure, would compress margins.

## The inventory tell — best single leading indicator

819 km of submarine cable inventory at end-2025 represents approximately 11 months of run-rate (annual delivery was 924 km). This is the single best leading indicator of whether the thesis is working or stalling.

**Bullish read:** deliveries about to ramp in 2026/27. The peak-season installation window (Q2/Q3) absorbs this inventory rapidly. Could drive +30–50% sequential revenue growth in the subsea segment.

**Bearish read:** working capital piling up because downstream installation is delayed. Replays the 2025 pattern — finished goods rising while revenue stays flat.

Which way it goes is observable in Q2 and Q3 2026 quarterly earnings. The KPI to watch: subsea cable kilometres delivered versus inventory days outstanding.

## Where Orient doesn't disclose — and how to triangulate

Six material gaps in standard disclosure. Acknowledging them proactively (with the triangulation workaround for each) is the difference between a sophisticated pitch and an exposed one.

| Metric not disclosed | How to triangulate |
|---|---|
| Customer concentration beyond top-5 | Cross-reference China public procurement tender awards with developer 10-K suppl |
| Per-project Average Selling Price | Reverse-engineer by dividing bid award values by route km from provincial bidding d |
| Quarterly submarine volume in km | Use inventory build as leading indicator + maritime AIS data tracking Dongfang Haig |
| Geographic export breakdown | Track named project deliveries: Inch Cape (UK), Hollandse Kust West Beta (NL), Ba |
| Capex and utilization by facility | "Construction in Progress" ledger discloses RMB 10.3 M for northern base; Yangjian |
| Contract cancellation penalty formulas | The "expected refund = RMB 0.00" disclosure functionally confirms non-refundability |

SECTION 11

# Dated catalyst calendar

Calendar events — not "should re-rate over time." Each event below is dated, named, and individually proves or breaks a piece of the thesis. The 12-month window through Q4 2027 has the highest density of inflection points.

| Window | Event | Signal |
|---|---|---|
| Q2 2026 | Netherlands 3.5 GW offshore wind tender launch (CfD framework) | + if Orient bids |
| Q2 2026 | Q2 earnings — subsea km delivered vs 819 km inventory benchmark | + if ↑ velocity |
| Q2 2026 | China bidding volume | – if <2 GW (tripwire 1) |
| Q3 2026 | UK Allocation Round 8 (AR8) — brought forward to July | + awards size |
| Q3 2026 | France AO10 award — 10 GW expected to land in fall 2026 | – 60% local content |
| Q3 2026 | Q3 earnings — peak delivery season | + if subsea +50% QoQ |
| Q4 2026 | Orient cumulative European orders expected to reach ~RMB 4 bn | + confirms 20% overseas |
| Q4 2026 | Yangjiang Sanshandao 500 kV HVDC commissioning | + first ±500 kV revenue |
| Q4 2026 | LS Cable BalWin5 FID — Korean competitor wins major European HVDC | – displacement risk |
| 1H 2027 | Germany CfD auctions resume after 2026 framework finalisation | + tender volume |
| 1H 2027 | Yantai (Shandong) facility online — RMB 3–4 bn projected output | + capacity step-up |
| 2H 2027 | Inch Cape full commercial operations begin — 1.1 GW UK project live | + track record proven |
| 2028 | FY27 EPS prints (UBS models RMB 3.38) — 18× P/E at current price | + re-rate |
| 2028 | ±800 kV HVDC R&D milestone — positions for trans-oceanic interconnectors | + moat extends |
| 2028 | LS Cable Tyne UK facility comes online — European supply gap starts closing | – Chinese window narrows |

**HOW TO USE THIS CALENDAR**

Pick the two or three events most relevant to your conviction. Set quarterly reminders. After each event, score it as thesis-confirming (+) or thesis-breaking (–). Two consecutive negatives = revisit the position size. Three or more negatives in a single quarter = the thesis is breaking, regardless of what the stock price does.

SECTION 12

# How to navigate the dashboard

The dashboard is structured top to bottom for reading in order. Here is what each section covers and the minimum you need from each.

| | |
|---|---|
| **Hero / KPIs** | Quick at-a-glance numbers. Skim and move on. |
| **Thesis** | Six bullet points distilling why this stock works. Start here. |
| **Cable 101** | Annotated diagram of what a submarine cable physically looks like. |
| **Voltage Ladder** | Why higher kilovolts = bigger moat. The technology hierarchy explained. |
| **Three Segments** | Detailed financial breakdown of submarine, land cable, and EPC services. |
| **Five Moat Pillars** | The five barriers to entry, scored individually. |
| **Vessel Fleet** | The cable-laying ships Orient owns — what makes the business defensible operationally. |
| **China Map** | Where in China the company manufactures vs where the demand is, province by province. |
| **Europe Pipeline** | European projects in the order book and pipeline, including the Morocco–UK XLinks cable route. |
| **Financials** | All historical and forecast numbers as interactive charts. |
| **Cost Drivers** | What moves the P&L (copper, segment mix shift). |
| **Valuation** | P/E analysis, peer comparison, scenario price targets. |
| **Bull vs Bear** | Interactive toggle showing both sides of the debate. |
| **Risk Matrix** | Risks ranked by probability and severity. |
| **Timeline** | Key catalysts from now through 2030. |

| | |
|---|---|
| **Floating Wind Option** | The long-term optionality not in the base case. |
| **Live Chart** | Real-time stock chart from TradingView. |

SECTION 13

# Glossary

Definitions for every jargon term that appears in the dashboard.

**Submarine cable** A power cable that lies on the seabed. Wraps copper conductors in insulation, lead, and steel.

**HVAC vs HVDC** High-Voltage Alternating Current vs Direct Current. HVAC is cheaper to make but loses more power over long distances. HVDC is harder to manufacture but works for very long distances (over 100 km). HVDC is what makes the Morocco–UK interconnection possible.

**500kV / ±525kV** Voltage ratings. Higher is harder to manufacture but lets you carry more power over longer distances. 500kV AC and ±525kV DC are the cutting edge today; only ~10 companies globally can produce them.

**XLPE** Cross-Linked Polyethylene — the plastic insulation used in modern submarine cables. The specific chemistry is part of what makes high-voltage cable manufacturing genuinely hard.

**Inter-array vs export cable** Inter-array cables connect individual turbines within a wind farm (lower voltage, more competitors). Export cables connect the wind farm's substation to the grid on land (higher voltage, fewer competitors, higher margin). Orient makes both.

**Dynamic cable** A cable used with floating wind turbines. It moves with the turbine in waves and currents, so it needs special fatigue-resistance engineering. A specialised future market that Orient is already producing for.

**EPC** Engineering, Procurement, Construction. The "turnkey" service model where the supplier handles everything from design through installation, rather than just selling components.

**CfD (Contracts for Difference)** A subsidy mechanism European governments use to guarantee wind farm developers a stable electricity price for 15–20 years. Removes price risk, makes financing easier, supports the long-term demand pipeline for cables.

**15th Five-Year Plan** China's national policy framework for 2026–2030. Targets 100 GW of cumulative offshore wind installed capacity by 2030.

**Order backlog** Contracts already signed but not yet delivered or recognized as revenue. Gives forward visibility on revenue and earnings.

**Gross margin** Revenue minus the direct cost of producing the product, expressed as a percentage. Higher = more pricing power and structural defensibility.

**Net cash** Cash on the balance sheet minus debt. Positive net cash means the company has no net leverage and can fund growth internally.

**P/E ratio** Price per share divided by earnings per share. Lower = cheaper. Used to compare valuations across companies and across time.

**ROIC** Return on Invested Capital — how much profit the company generates per dollar of capital invested in the business. Higher = better business quality.

**Oligopoly** A market dominated by a small number of firms. China's submarine cable market is a 3-player oligopoly with 87% combined share.

**Inch Cape** A large UK offshore wind project off the east coast of Scotland. Orient supplies RMB 1.8 billion of cable for it; grid connection scheduled for 2027.

**XLinks** A proposed 4,000 km HVDC cable that would carry Moroccan solar power to the UK. Orient owns 2.4% of the developer.

**Fanshi I & II** Major Chinese offshore wind projects developed by CGN. Orient supplies RMB 1.7 billion of cable; grid connection in 2025/2026.

**Dongfang Haigong** The brand name for Orient's fleet of three cable-laying and transport vessels. DHG-07 is the 18,000-tonne platform that ships completed cables to Europe.

---

*End of pre-read. The dashboard at **juankhye.github.io/ningbo-orient-603606-dashboard** contains the supporting charts, financial tables, maps, and live stock data referenced above.*